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MILITARY STANDARD

FASTENER TEST METHODS

METHOD 19

FASTENER SEALING

AMSC N/A

FSC 53GP

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FORWARD

1. This military standard is approved for use by all Departments and Agencies of the Department of Defense.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (SESD), Code 53, Lakehurst, NJ 08733-5100, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

3. This standard sets forth a standard test method for determining the ability of a fastener to restrict fluid (liquid or gas) leakages.

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1. SCOPE

1.1 Applicability. This method covers the procedures and apparatus for determining a fastener's ability to restrict leakage of a fluid (liquid or gas) medium, under static conditions only. These fasteners are not externally or dynamically loaded.

2. REFERENCED DOCUMENTS

Not applicable.

3. DEFINITIONS

Not applicable.

4. GENERAL REQUIREMENTS

4.1 Test apparatus.

4.1.1 Pressure pot. A typical pressure pot is illustrated on Figure 1. The pressure pot shall be connected to a suitable pressure source so that the test conditions of Section 5 are accomplished.

4.2 Test specimen.

4.2.1 Test plate. The test plate to be used in conjunction with the pressure pot shall be in accordance with Figure 2; when fasteners with a diameter larger than 0.250-inch are to be tested, the number of specimens installed in each plate may be reduced. Grooved spacers which provide a leak path to the fastener shank shall be installed on the pressure side of the plate to provide for different grip lengths.

4.2.2 Fastener holes. Fastener holes shall be drilled perpendicular to the test plate surface within 1 degree. When countersinking is required, the countersink shall be concentric with the hole within 0.002 FIM, and the depth shall be such that the installed fastener is flush within +0.005 or -0.000-inch. Unless otherwise specified in the product specification, hole diameters used in the preparation of the test plate shall be in accordance with Table I.

4.2.3 Assembly. All solid rivets, swaged collar fasteners, and blind fasteners shall be installed to meet the appropriate product specification requirements. Sealing type nuts shall be torqued to the minimum specified value for the particular nut-bolt combination. Unless otherwise specified, lubrication, except as part of the product specification, shall not be applied.

4.2.3.1 Spacer usage. A spacer will be used to create leak paths to the fastener shank under test. When a sealing head or shank is under test, spacers will be placed under the nut, and the nut side pressurized. When a sealing nut is being tested, the spacer shall be used under the head of the mating bolt, and the pressure applied from the bolt head side. For installation formed fasteners, no spacer will be required.

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4.2.4 Surface finish.

4.2.5 Cleaning.

TABLE I. Hole limits (diameter).

Diameter	NAS 618 Shank Type Fasteners		Solid Rivets		Sealing Nuts	
Nominal Size (Inch)	Min.	Max.	Min.	Max.	Min.	Max.
0.138	0.138					
0.164	0.164	0.168	---	---	0.167	0.171
0.190	0.190	0.194	0.193	0.197	0.193	0.197
0.250	0.250	0.254	0.253	0.257	0.253	0.257
0.313	0.312	0.316	0.316	0.320	0.316	0.320
0.375	0.375	0.379	0.378	0.382	0.378	0.382
0.438	0.438	0.442	0.442	0.446	0.441	0.445
0.500	0.500	0.505	0.504	0.509	0.503	0.507
0.563	0.562	0.568			0.566	0.570
0.625	0.625	0.630	-----	-----	0.628	0.632
0.750	0.750	0.757			0.753	0.757
0.875	0.875	0.882	-----	-----	0.878	0.882
1.000	1.000	1.010			1.003	1.007
1.125	1.125	1.135	-----	-----	1.128	1.132
1.250	1.250	1.260			1.253	1.257
1.375	1.375	1.385	-----	-----	1.378	1.382
1.500	1.500	1.510			1.503	1.507

5. DETAIL REQUIREMENTS

5.1 Test procedures.

5.1.1 Leakage test. The test plate, Figure 2, shall be installed in the pressure pot, Figure 1, and the upper flange secured to the lower flange face. A typical installation is shown on Figure 3. The pressure pot shall be filled with the test medium as specified in the product specification. A fluorescent U.V. type dye shall be added to the test medium to permit detection of possible leakage. The test plate shall be cycled between the specified test pressure limits. The test medium temperature shall be maintained within 20°F throughout the test. The cyclic speed shall be 25-50 cycles per minute. When a gas medium is utilized, the plate shall be covered with a liquid which will not react with the gas medium.

5.1.1.1 Leak path verification. When testing sealing nuts, pre- and post-test leak path verification test cycles shall be performed. The pre-test verification cycle shall consist of cycling the pressure pot in accordance with the product specification before fully seating and torquing the nuts to be tested. The test medium shall be observed to leak through to the top of the test plate. After completion of the leakage test of the sealing nut under

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evaluation, and with the pressure pot still holding the maximum test pressure, the tested nuts shall be loosened and the test medium shall again visibly leak through the top of the test plate. This will be the post-test leak path verification cycle. Safety precautions shall be observed in conducting the leak path verification tests, including a cover or encasement of the pressure pot.

5.1.2 Failure. Unless otherwise specified, the test shall be conducted to the pressure designated in the prodict specification. Failure shall be defined as visible leakage exhibited by the presence of the liquid medium on the test plate detected by means of an ultra violet lamp, or bubbles of the gas medium in the liquid covering the test plate.

6. NOTES

6.1 Test report. The test report shall contain the following data:

a. Fastener description.

1. Part number and cage code.

2. Lot identification.

3. Material.

4. Heat treat.

5. Grip length.

6. Mating part.

7. Measured fastener diameter.

b. Test machine.

1. Model and serial number.

2. Calibration date.

3. Test plate hole diameter.

c. Installation procedure.

d. Test method (where options are available).

e. Test pressure and time cycle.

f. Test duration.

g. Type of failure.

h. Sheet material description.

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i. Cause and length of interruption during test.

j. Results of all inspection.

k. Specimen preparation.

l. Test medium.

m. Plate deflection.

n. Results of pre-test leak path verification cycle (sealing nuts).

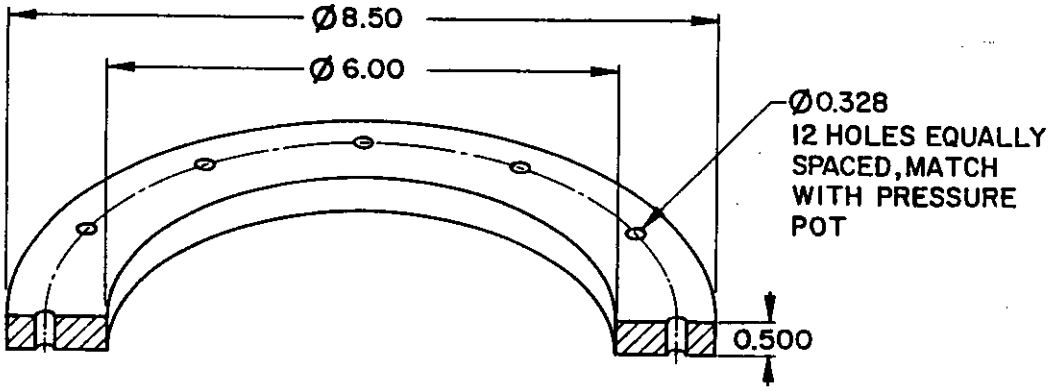
o. Results of post-test leak path verification cycle (sealing nuts).

Custodians:
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Navy - AS
Air Force -99

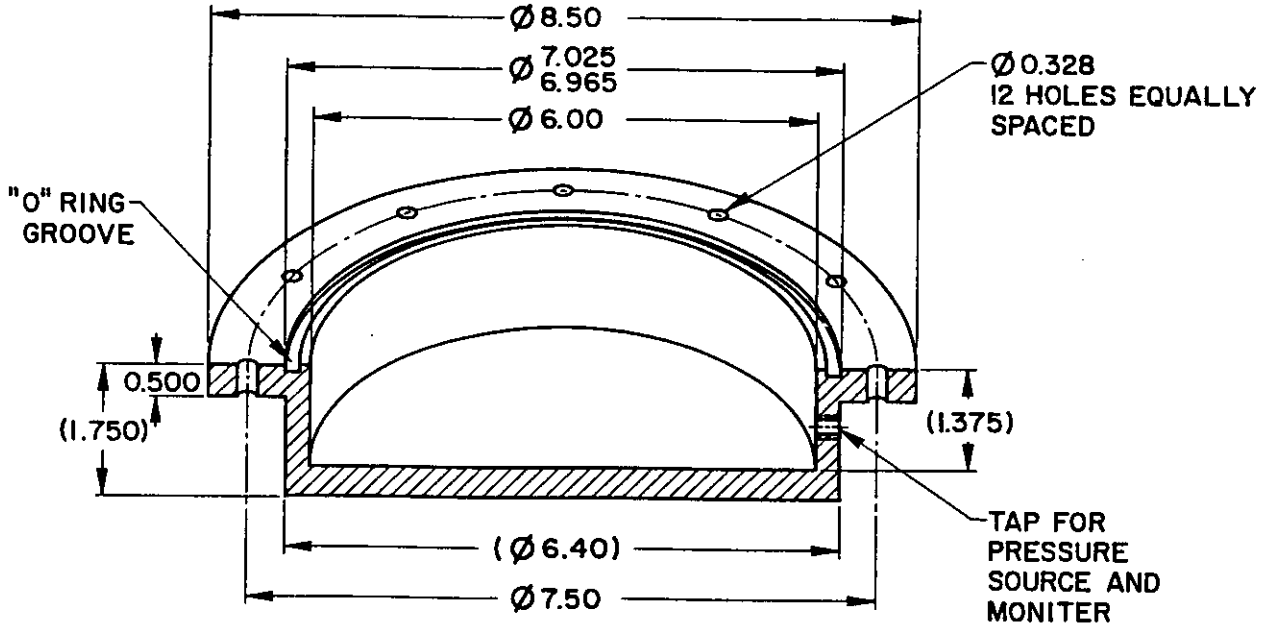
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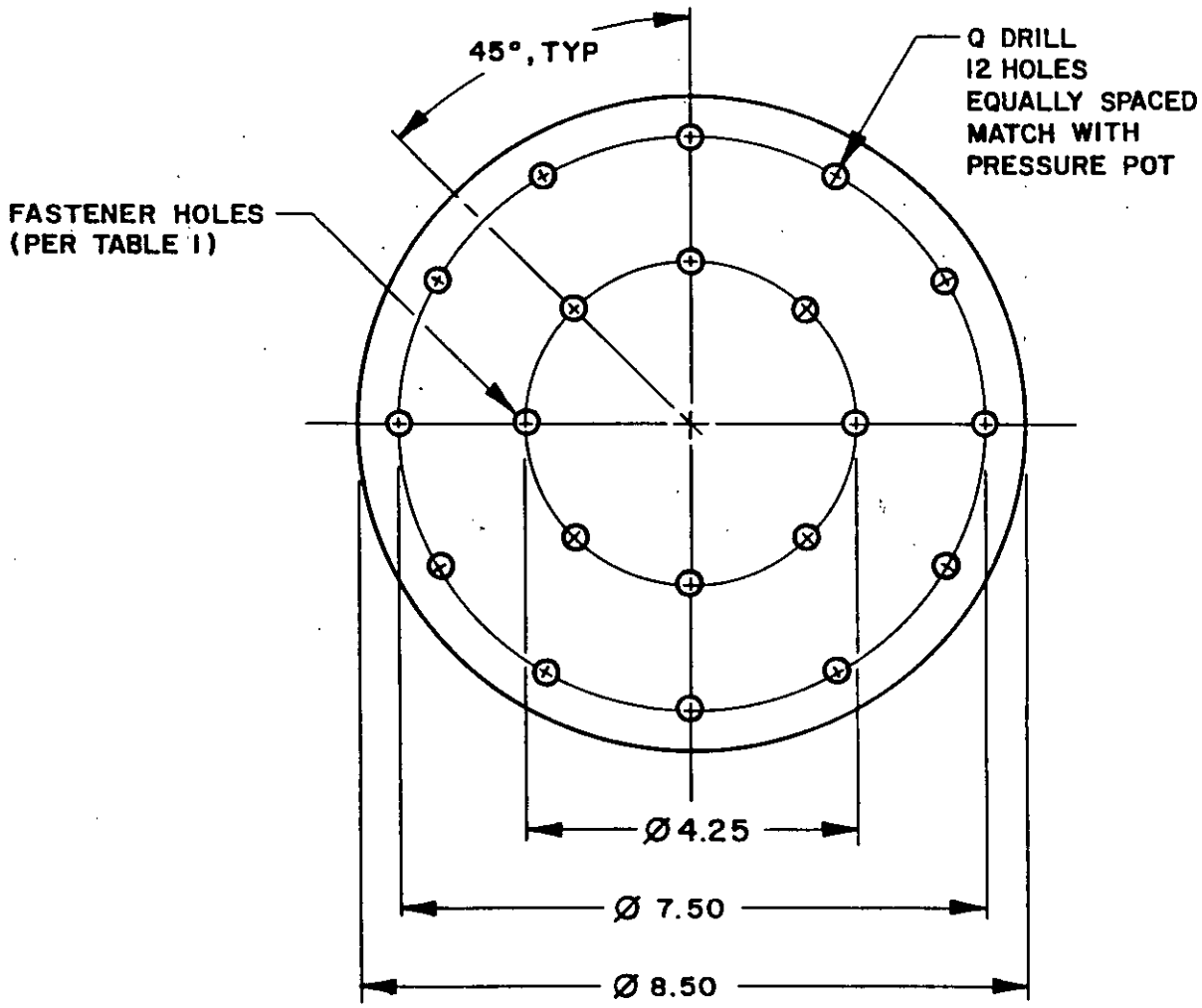
UPPER FLANGE

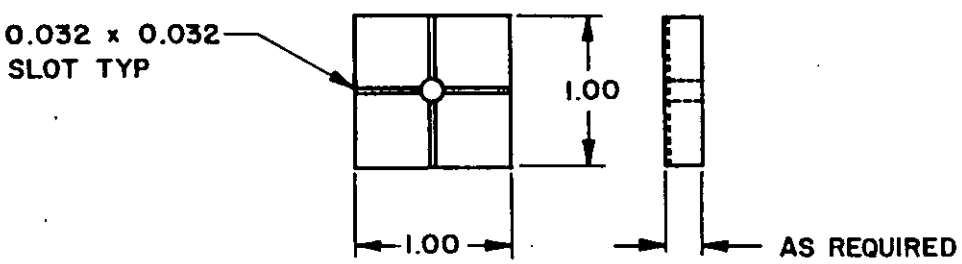


DIMENSIONS IN INCHES
MATERIAL: STEEL
TOLERANCES: .XX ± 0.01
.XXX ± 0.005

FIGURE 1. Pressure pot.

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DIMENSIONS IN INCHES
MATERIAL: 2024-T3 ALUMINUM PLATE 0.125 THICK EXCEPT AS NOTED
TOLERANCES: DECIMALS .XX ± .02
.XXX ± .005

FIGURE 2. Test plate and spacer.

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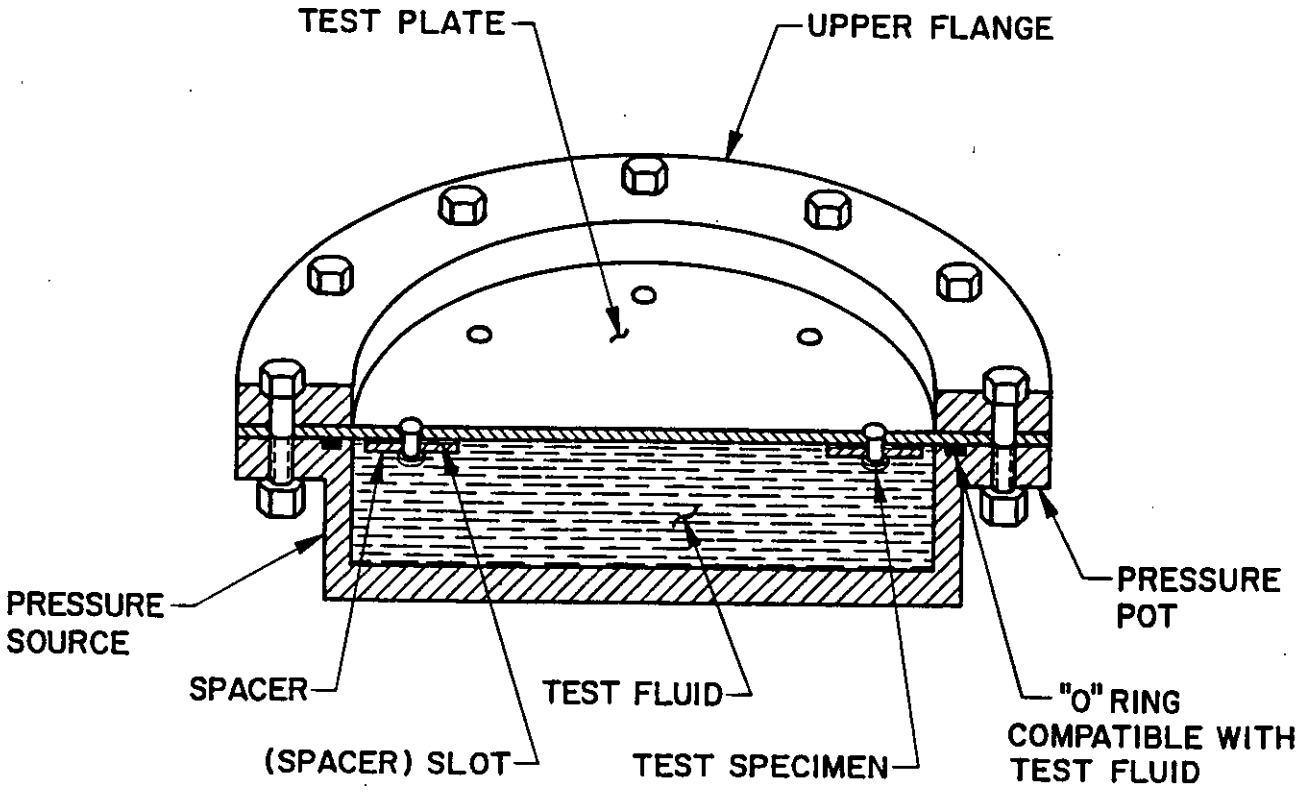


FIGURE 3. Test installation.

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